

An omnidirectional asymmetric mobile robot for narrow aisle spaces

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Abstract—The constrained spaces in narrow aisles of air-planes and manufacturing facilities limit the use of mobile robots. Utilizing Mecanum wheels allows mobile robots to move smoothly in any direction without needing to turn, enhancing maneuverability and efficiency in constrained environments. However, the sizeable Mecanum wheel and the individual motor assembly required for independent control restrict its use in mobile robot designs for maneuvering in narrow spaces. This work proposes an asymmetric design to overcome the challenge of using Mecanum wheels in small-size robots and analyses the kinematic requirements due to asymmetric design. Numerical simulation and experimental tests confirm the platform's ability to navigate tight spaces, make sharp turns, and carry various payloads, significantly improving operational efficiency. The proposed asymmetric mobile robot offers a practical solution for enhancing the performance of automated guided vehicles (AGVs) and robotic systems in space-limited settings.

I. INTRODUCTION

An omnidirectional mobile robot enables a vehicle [1] or robotic platform to move freely in any direction—forward, backward, sideways, and diagonally—using specialized wheels like omni-wheels or Mecanum wheels [2], [3], each powered by an independent motor. This advanced design, controlled by sophisticated systems and sensors, allows for precise navigation and obstacle avoidance, making it ideal for maneuvering in confined and dynamic environments such as warehouses [4], retail stores, manufacturing plants, and healthcare facilities [5]–[7]. By maximizing maneuverability and efficiency, an omnidirectional robot significantly enhances operational capabilities and space utilization across various industries [8].

The conventional design of the mobile robot [9], [10] with wheels on either side in a collinear fashion requires the size of the robot to be wide enough to accommodate two motors and wheel hubs in a single line if the wheels are driven independently [11]. This hinders the reduction of the robot's width and poses a problem for the applications targeted in very narrow aisle spaces [12]. Also, as the torque requirement increases for higher load-bearing capacities, the increase in motor sizes for higher torques presents a more

significant challenge while designing a compact mobile robot [13], [14]. Avoiding the collinear placement of motors while placing the wheels on either side in the same axis requires the designers to resort to complex drive train mechanisms (synchronous belts and drive wheels) with asymmetric placement of motors, which may result in further erroneous movements of the robot [15], [16]. Further, the conventional designs with regular wheels suffer additional space requirements for a dedicated steering mechanism for locomotion [17], [18], which may hamper the compactness of the platform, making it unfit for applications in narrow spaces [19], [20].

This paper proposes a compact mobile robot design tailored for autonomous and semi-autonomous platforms, specifically engineered to navigate through narrow and congested spaces. The main challenge of maneuvering in narrow space through contradicting options of *omnidirectional maneuverability using Mecanum wheels* and bulky size is addressed through an asymmetric design. Further, this work contributes in analysing the maneuverability of proposed asymmetric design by developing kinematic model and and prototyping an experimental set-up. Extensive testing results are presented on repeatability of desired maneuvers through extensive simulations and experiments.

The next section presents an intricate breakdown of the proposed design of omnidirectional mobile robot, outlining its unique features and configurations. Additionally, the section III presents the derived kinematic equations for the proposed mobile robot design along with graphical representations for better understanding. To validate the efficacy of the proposed design, numerical simulations are presented in section IV, comparing the predicted performance against established benchmarks. Finally, the *Prototype Development and Testing* section showcases the physical realization of the proposed model, along with rigorous testing procedures to validate its functionality and performance in real-world scenarios. This comprehensive approach ensures a thorough exploration and validation of the novel mobile robot design for smooth locomotion in narrow spaces.

II. PROPOSED OMNIDIRECTIONAL ASYMMETRIC MOBILE ROBOT

The proposed mobile robot is constructed with an alternate placement of drive wheels such that the lower bound for the mobile robot width is limited by the length of one motor and one wheel hub, as shown in Figure 1. In this figure, the front side of the robot platform is towards the right side of this paper. It is worth noting that the right and left wheels at both the front and rear ends of the mobile robot have parallelly displaced axes of rotation. This arrangement reduces the

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Fig. 1: Top view of the chassis for the proposed robot with closely placed lateral wheels, Variant 1.

mobile robot width to the combined length of one motor and one wheel hub. Mecanum wheels are used for this novel arrangement as they provide omnidirectional movements for the robot platform. Thus, the novel mobile robot can perform holonomic movements, i.e., the platform can control the movements along all the degrees of freedom along the ground plane. This novel alternate arrangement of motorized drive wheels will reduce the width of the mobile robot to half compared to the conventional/traditional mobile robot designs with the collinear arrangement of drive wheels already available. Further, as the mobile robot width is dictated only by the size of a single motor-wheel-hub assembly, requirements for larger motors for higher torque requirements and much higher load-bearing capabilities can be easily accommodated on the proposed mobile robot with minimal impact on suitability for narrow aisle applications. Thus, the achieved half-width narrow mobile robot design will facilitate the locomotion of the robot platform in very narrow and congested spaces such as aisles of airplane cabins, railway coaches, congested warehouses, seminar/conference rooms, or other spaces with narrow aisles. Further, the omnidirectional movements facilitate higher maneuverability and can plan the motion smoothly for the platform along tight, narrow spaces. In Figure 1 the wheels are shown to be placed towards the longitudinal ends of the robot, and the left-right pair of wheels are quite close to each other. This design is hereafter referred to as ‘variant 1’ throughout the remainder of this paper. This geometry of wheels was chosen to ensure the platform’s stability without any other support wheels. Another variant of the design is shown in the Figure 2, referred as ‘variant 2’. In this

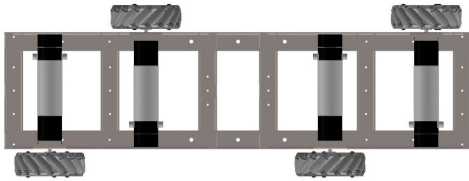


Fig. 2: Top view of the chassis for the proposed robot with evenly distributed lateral wheels, Variant 2.

alternate design, the wheels are evenly distributed along the length of the robot. However, this variant lack stability as the wheels are not evenly distributed laterally and need additional arrangements to stabilize the platform. Attaching four non-motorized support wheels or caster wheels (not depicted in the figure) at the lateral opposite end of each

drive wheel enhances stability and increases the load-bearing capacity of this design variant. Understanding the movement of the robot and designing control algorithms for navigation requires deriving the robot’s kinematic model. The next section discusses the kinematics of the robot platform.

III. KINEMATIC MODELING

A top view of the proposed mobile robot is depicted in the diagram 3. The large rectangular box depicts the proposed mobile robot with four wheels, shown with the shaded rectangles at the corners. It is worth noting that the tilting pattern on the wheels is maintained identically to the roller axis on the Mecanum wheels for this configuration. The robot pose (x, y, θ) is defined with respect to the inertial frame I as shown in the figure. The mobile robot’s body frame, B, is shown in the diagram, and the origin ‘ O_b ’ is located at the geometric center of the robot platform. The ‘ x_b ’ axis of the body frame is aligned with the robot’s longitudinal surge velocity direction (forward direction), and ‘ y_b ’ is aligned towards the robot’s left lateral direction. As

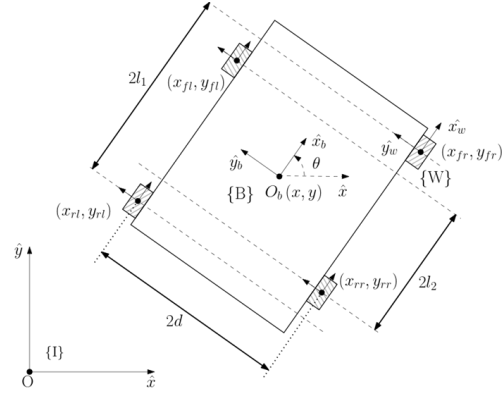


Fig. 3: Schematic diagram of kinematic model.

discussed above, unlike the conventional mobile robot, the wheels at the left and right of the platform at both front and rear ends are noncollinear. The axes of rotation for all the wheels, illustrated with dashed lines in 3, are parallelly displaced. Also, a wheel frame W, is shown at the center of each wheel, and the wheel origins are marked with their x and y coordinates with respect to the body frame, B, with two subscripts indicating their position on the mobile robot. The first subscript, ‘ f/r ’ indicates front or rear, and the second one, ‘ l/r ’, indicates the lateral position, left or right, on the mobile robot. The wheel frame W axes are parallel to the body frame B such that ‘ x_w ’ and ‘ y_w ’ are aligned with ‘ x_b ’ and ‘ y_b ’ respectively. The velocity vector of the robot is defined as $\tilde{\mathbf{v}} = [u, v, r]^T$ where ‘ u ’ denotes the surge velocity along the ‘ x_b ’ axis, ‘ v ’ is the lateral velocity along the ‘ y_b ’ axis, and ‘ r ’ is the angular velocity with which the robot turns around about the ‘ z_b ’ axis (not shown in the figure) which pointed outside of the paper. The angular velocity vector of the wheels is denoted with ‘ $\tilde{\boldsymbol{\omega}} = [\omega_{fr}, \omega_{fl}, \omega_{rr}, \omega_{rl}]^T$ ’ where the individual wheel’s angular velocity is denoted by ω with subscripts for longitudinal and

lateral positions on the mobile robot, similar to wheel origins. The positive rotation direction is considered with respect to 'y_w' axis; thus, the wheel's rotation in the forward direction of the mobile robot is considered positive irrespective of the wheel's position. In this novel arrangement, all the wheels considered are of the same size with radius 'a.' The wheel origins are represented using fewer variables for further analysis, accounting for the symmetry of the wheels' placement along the longitudinal axis of the robot platform. The distance between the frontmost and the rearmost wheel is 2l₁ along the x_b axis, and the distance between the two middle wheels is 2l₂, as shown in the figure. Also, the distance between the left and right wheel centers is 2d along the y_b axis. Thus, the coordinates of the front right wheel (x_{fr}, y_{fr}) is (l₁, -d) and the front left wheel (x_{fl}, y_{fl}) is (l₂, d) about the body frame. Then the rear wheels coordinates are (x_{rr}, y_{rr}) = (-l₂, -d) and (x_{rl}, y_{rl}) = (-l₁, d). Now, the kinematics of the robot in the 2D plane is discussed below based on the above setting. The inverse kinematics of the mobile robot that calculates individual wheel angular velocities from the linear and angular velocities of the mobile robot is described by equation 1.

$$\tilde{\mathbf{w}} = \frac{1}{a} M \tilde{\mathbf{v}}, \quad M = \begin{bmatrix} 1 & 1 & (l_1 + d) \\ 1 & -1 & -(l_2 + d) \\ 1 & -1 & (l_2 + d) \\ 1 & 1 & -(l_1 + d) \end{bmatrix} \quad (1)$$

The above equations map the velocities of the mobile robot in the body frame B to individual wheel velocities in W frame. It is worth noting that during actual implementation, it may be required to perform one more mapping from the W frame to the individual motor frame as the motor axis along the left and right sides are aligned in opposite directions. Similarly, forward kinematics calculates the robot platform velocity from the individual wheel angular velocities. This model is required to estimate the robot's position and velocities during control implementations. The equation for forward kinematics is given below.

$$\tilde{\mathbf{v}} = \frac{a}{4} M^\dagger \tilde{\mathbf{w}}, \quad (2)$$

$$M^\dagger = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & -1 & 1 \\ \frac{2}{l_1+l_2+2d} & \frac{-2}{l_1+l_2+2d} & \frac{2}{l_1+l_2+2d} & \frac{-2}{l_1+l_2+2d} \end{bmatrix} \quad (3)$$

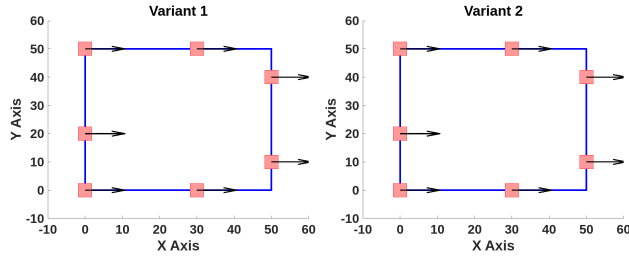
Further, the dynamics of the robot pose with respect to the body frame velocities can be calculated using the following set of equations.

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \tilde{\mathbf{v}} \quad (4)$$

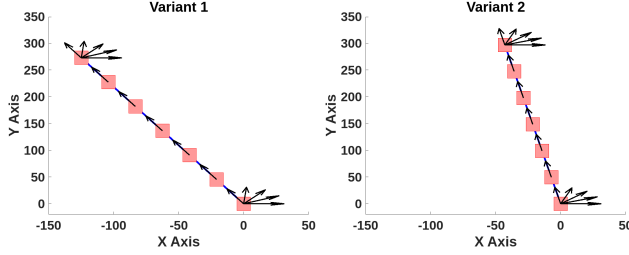
The above set of equations completely describes the kinematics of the novel mobile robot. The set of equations is verified through numerical simulations, which will be discussed in the next section.

IV. NUMERICAL SIMULATION

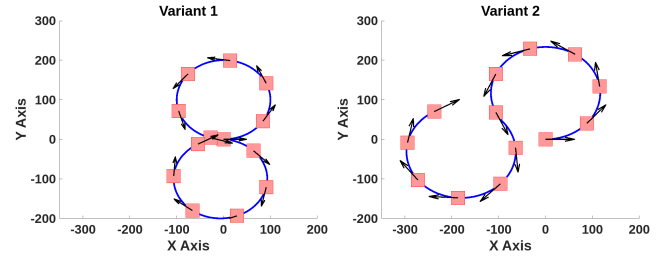
The numerical simulations of the kinematic equations are performed in MATLAB to validate the correctness of the equations for both the proposed robot variants. The robot model parameters considered for the variant 1 are l₁=15, l₂=10, d=5 and a=1. All the model parameters for variant 2 is same as variant 1 except the value of l₂=5. The simulations are performed with the robot's initial pose about the inertial frame I to be (x = 0, y = 0, θ = 0). Four different sets of simulations are performed for this initial condition for varying robot body frame velocities for both the variants. In each of these simulations, the wheel angular velocities are calculated from the body frame velocities using equations given in Equation 1. Further, the equations in Equation 1 and Equation 2 are integrated using ODE45 to calculate the robot positions plotted in Figure 4. In all the cases shown in Figure 4, the robot's path is shown using the blue line. Also, the robot's position is shown at some intervals with a red square marker, and the robot's heading for those positions is depicted with a black arrow on the marker. In each of the cases shown in Figure 4, the path traced for variant 1 and variant 2 is depicted in the left and right sides respectively. In case 1, the proposed mobile robot is made to move in a square shape where, initially, the velocities are set to (u = 10, v = 0, r = 0) for the first 5 seconds. Once the mobile robot moves forward 50m, the velocity is modified to (u = 0, v = 10, r = 0) and maintained for the next 5s, which makes the robot move 50m in the 'y_b' direction. Further, velocities are set to (u = -10, v = 0, r = 0) and (u = 0, v = -10, r = 0) for the subsequent 5s intervals to complete the square. The corresponding wheel angular velocities calculated are (ω_{fr} = 10, ω_{fl} = 10, ω_{rr} = 10, ω_{rl} = 10), (ω_{fr} = 10, ω_{fl} = -10, ω_{rr} = -10, ω_{rl} = 10), (ω_{fr} = -10, ω_{fl} = -10, ω_{rr} = -10, ω_{rl} = -10), and (ω_{fr} = -10, ω_{fl} = 10, ω_{rr} = 10, ω_{rl} = -10) respectively. The path taken by the robot and the heading, as detailed above, is shown in Figure 4(a). This case verifies the forward/reverse and lateral movements for the mobile robot kinematic equations. In case 2, the mobile robot is made to move with velocity (u = 10, v = 0, r = 0.1) for half of the time, and later, the velocity is modified to (u = 10, v = 0, r = -0.1) for the next half. The wheel angular velocities calculated for these rotations are (ω_{fr} = 12, ω_{fl} = 8.5, ω_{rr} = 11.5, ω_{rl} = 8) and (ω_{fr} = 8, ω_{fl} = 11.5, ω_{rr} = 8.5, ω_{rl} = 12) respectively. The intention is to check the angular rotations of the mobile robot while moving forward. The first set of velocities should track a circle while moving in a counterclockwise direction, while in the second half, it tracks a circular path in a clockwise direction. The path taken by the robot is shown in Figure 4(b), which verifies the kinematics. In case 3, pure rotation of the mobile robot is validated by giving (u = 0, v = 0, r = 0.1) for performing counterclockwise rotation. Then the robot is moved forward with (u = 0, v = 10, r = 0), and the robot is provided with negative angular velocity (u = 0, v = 0, r = -0.1) to rotate in a clockwise direction.



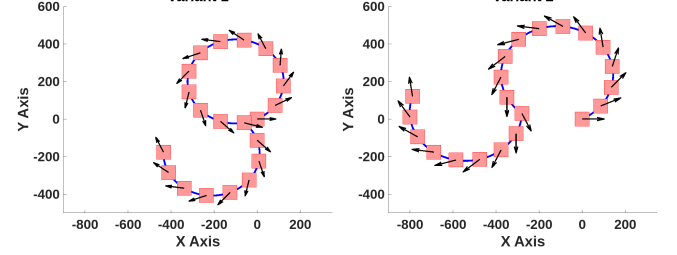
(a) Simulation for tracing a square; this simulation tests the lateral and longitudinal movement capabilities for the kinematics.



(c) Illustration of pure rotation of the robot kinematics.



(b) Simulation for tracing circular loops in both clockwise and anticlockwise directions.



(d) Simulation for simultaneous application of constant non-zero velocities for u , v , and r . Sign of r is reversed in the second half.

Fig. 4: Numerical simulation of different maneuvers for the proposed kinematics for Variant 1 (left) and Variant 2 (right)

The wheel angular velocities for these three segments are $(\omega_{fr} = 2, \omega_{fl} = -1.5, \omega_{rr} = 1.5, \omega_{rl} = -2)$, $(\omega_{fr} = 10, \omega_{fl} = 10, \omega_{rr} = 10, \omega_{rl} = 10)$ and $(\omega_{fr} = -2, \omega_{fl} = 1.5, \omega_{rr} = -1.5, \omega_{rl} = 2)$ respectively. The robot is moved from the position after the counterclockwise rotation to show both the rotations clearly and distinctly. This case is shown in Figure 4(c), where the robot initially turns counterclockwise and then moves forward to reach a distinct point where it rotates in the clockwise direction. Now, in case 4, the mobile robot is directed to move with velocities $(u = 20, v = 10, r = 0.1)$ for the first half, and velocities are modified to $(u = 20, v = -10, r = -0.1)$ in the second half. The angular velocities calculated for the wheels in this case is $(\omega_{fr} = 32, \omega_{fl} = 8.5, \omega_{rr} = 11.5, \omega_{rl} = 28)$ and $(\omega_{fr} = 8, \omega_{fl} = 31.5, \omega_{rr} = 28.5, \omega_{rl} = 12)$ respectively, which are feasible inputs for the motors. The resultant path is plotted in Figure 4(d). It can be observed that, in the first half, the robot moves from its initial position in a counterclockwise direction while moving laterally towards the left side of the mobile robot. This is evident from the robot heading not being aligned with the path. Further, the motion is clockwise for the second half while laterally moving towards the right side. The numerical simulations for all these different cases depict the movement of the mobile robot after calculating robot positions by integrating the forward kinematics. These results verify the above-derived kinematic equations for the novel mobile robot design. Also, these simulations illustrate a few possible maneuvers with the proposed mobile robot. Furthermore, it is important to observe that variant 1 achieves a faster orientation change compared to variant 2 at the same angular velocity, as demonstrated in Figure 4(c). A prototype of the proposed mobile robot was developed to validate the design further. The next section discusses the details of the

prototype developed to test the proof of concept.

V. PROTOTYPE DEVELOPMENT AND TESTING

A prototype of the proposed design (variant 2) was developed for further validation and testing. The chassis for the mobile robot was laser cut on a high-density acrylic sheet with a thickness of 5mm. The chassis length is 260mm, and the width without the wheels is 77mm. In the prototype, the motor mounts were uniformly spaced along the longitudinal axis, as shown in Figure 5. Four mecanum wheels connected

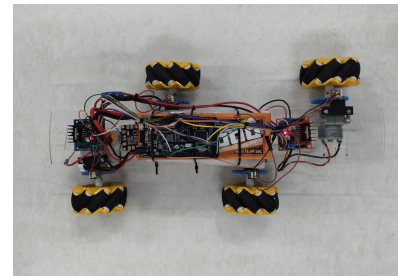


Fig. 5: Prototype of the mobile robot

in the alternate fashion are actuated by 12V geared DC motors with 30 rpm. The distance parameters for the wheels l_1 , l_2 , and d are 165mm, 55mm, and 87.5mm, respectively. The wheels of the mobile robot are equipped with a coarse wheel encoder with 36 pulses per revolution (PPR) for odometry. The motors are powered by an 11.1V Lithium polymer battery through a PDB-XT60 with BEC (Battery Eliminator Circuit) 5V and 12V Power Distribution Board via two L298N dual motor driver modules. The battery is tightly attached to the body using velcro loops. The mobile robot is controlled via an Arduino Mega board, which receives the user setpoint inputs from the HTML web interface provided

by ESP32 of the Arduino Mega board. The user can connect to the same wireless network and access the webpage to see the remote interface for controlling the robot. A PID controller is implemented in the firmware, which reads the encoder measurements to calculate the wheel velocities and regulate the error between the provided setpoint and the measured speed. The inverse kinematics Matrix for the robot with the above dimensions is given in equation 5.

$$M = \begin{bmatrix} 1 & -1 & -246 \\ 1 & 1 & 176 \\ 1 & 1 & -176 \\ 1 & -1 & 246 \end{bmatrix} \quad (5)$$

The proposed mobile robot is designed for moving in narrow aisle environments, which demands the robot to have excellent longitudinal and lateral movement characteristics. A thorough evaluation of the longitudinal and lateral movements of the prototype was done experimentally. Four sets of experiments were conducted to evaluate the performance of the designed prototype where the robot is moved in forward, reverse, left, and right directions. In each set, the robot is initially kept at the origin with a yaw angle of zero with respect to the inertial frame and is repetitively moved in one of the directions for five instances or runs to see the deviations from the desired path. The actual path taken by the robot is received from the Vicon motion capture system. The mean and standard error (SE) of the runs was calculated to evaluate the performance of the robot. The top view of the experimental setup is shown in Figure 6 and the relevant data for these experiments are shown in Figure 7. The



Fig. 6: Top view of the experimental setup

paths the mobile robot takes for the first case, where the robot is moved forward repeatedly for five runs, are given in Figure 7(a). The path taken by the robot for each run is shown in a different color. The velocity setpoints given to the robot are $(u = 100, v = 0, r = 0)$. The deviations from the desired path are depicted in Figure 7(b). The color scheme used for depicting the path and deviations from the path for each run is the same for all the figures. It can be observed that the maximum deviation of the mean is 3.65cm at a distance of 1.43m. The paths taken by the robot for the reverse maneuver experiment are provided in Figure 7(c), and the deviations from the desired path are given in Figure 7(d). Here, the velocity setpoints given to the robot are $(u = -100, v = 0, r = 0)$, and the maximum deviation of the mean from the desired path is 6.23cm at a distance of 2.46m, slightly higher compared to the forward

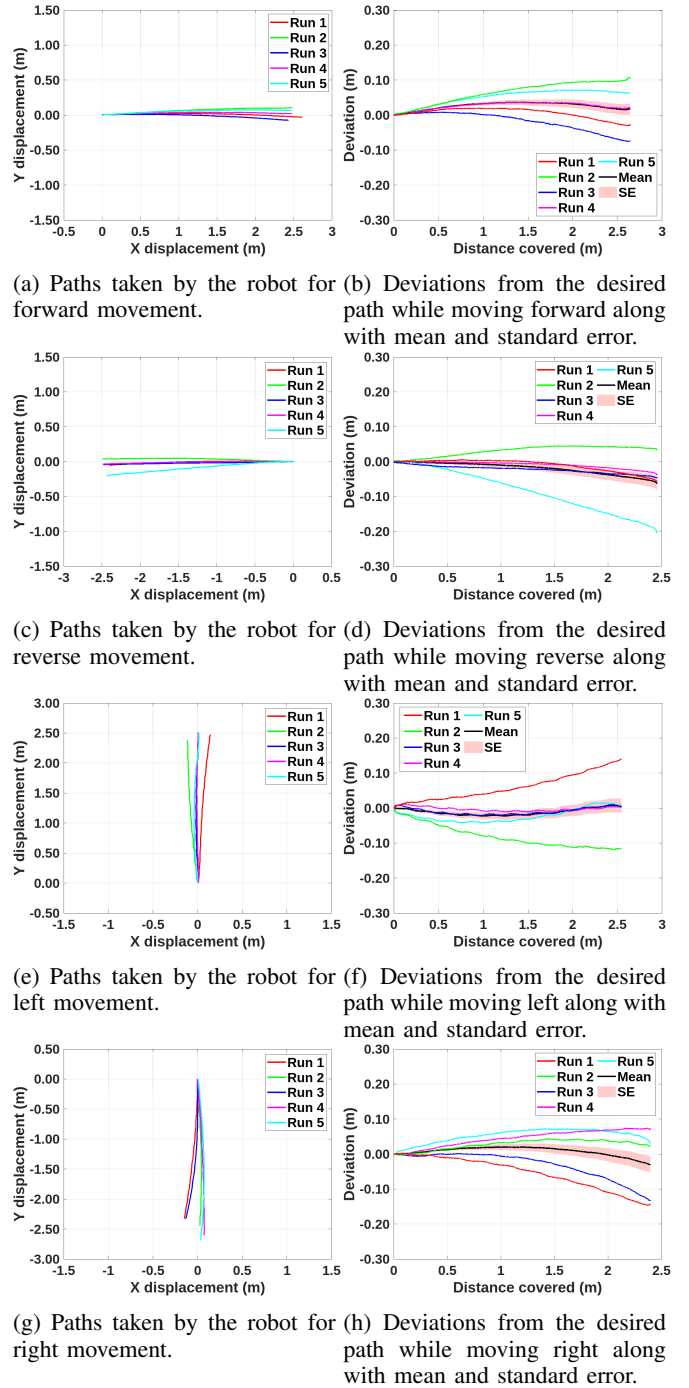


Fig. 7: Experimental data for the longitudinal and lateral maneuvers of the prototype.

movement. Figures 7(e) and 7(f) respectively show the paths taken by the robot and the corresponding deviations from the desired path while strafing the mobile robot toward the left direction. The velocity setpoints given to the robot are $(u = 0, v = 100, r = 0)$. In this case, the maximum deviation of the mean is approximately 2.28cm at a distance of 1.01m. Similar results for strafing the designed mobile robot towards the right are shown in Figures 7(g) and 7(h). The maximum deviation of the mean is approximately 3cm at a distance of 2.39m.

Even though the preliminary prototype design gives rea-

sonable performance, the actual implementation for narrow aisle spaces requires improved and better hardware. These results can certainly be improved by equipping a high-resolution encoder for the wheels compared to 36 PPR for the current design. Also, in this initial prototype, the wheels are connected to the motors through a 3D-printed coupling, which does not provide a stable orientation for the wheels, especially while strafing right and left. This effect can be evidently seen while comparing the deviation graphs for the experiments for longitudinal and lateral movements. Similarly, the temporary design of the battery fixture on the robot also accounts for a shift in the center of mass for the robot, which can affect moving the robot between each experiment. There is significant potential for future research to further investigate the stability of the platform by conducting a comparative analysis between the proposed and conventional chassis models. Moreover, numerical simulations of the proposed model, with variations in chassis parameters l_1 and l_2 , could be employed to determine the optimal design configuration. Another major issue that was noticed during the testing was the platform's instability in carrying loads due to the non-uniform placement of wheels around the corners of the robot. As discussed in the initial design description, this problem can be rectified by mounting castor wheels at the laterally opposite end of individual Mecanum wheels. Thus, the results from the experiments reinforce the feasibility of the proposed mobile robot for its use in narrow aisle spaces and demonstrate the performance of the prototype for different maneuvers.

VI. CONCLUSION

The omnidirectional mobile robot design effectively addresses the limitations of conventional chassis designs by enabling movement in lateral and longitudinal directions, offering distinct advantages in navigating narrow spaces. The derived kinematic model and results show the ability to move freely in any direction—forward, backward, sideways, and diagonally—enhances versatility and maneuverability. By reducing the mobile robot width to the combined length of one motor and one wheel hub, the design becomes exceptionally narrow, facilitating locomotion in congested environments such as airplane cabins, railway coaches, warehouses, and seminar rooms with tight aisles. Moreover, the omnidirectional movements enable smoother motion planning, ensuring seamless navigation along narrow and congested pathways. This comprehensive approach enhances the efficiency and adaptability of robotic platforms in challenging operational environments.

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